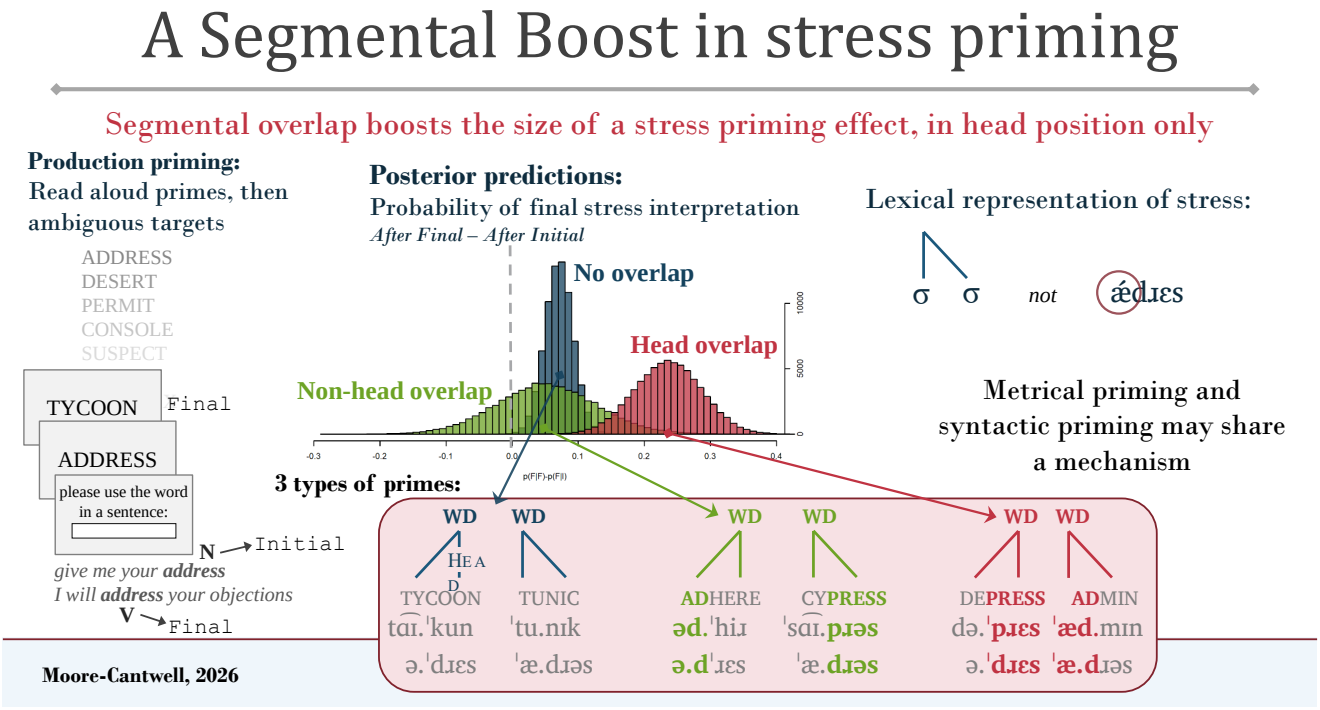
Graphical Abstract

A ‘segmental boost’ in production priming of stress

C. Moore-Cantwell



Highlights

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- Metrical structure is represented abstractly in the lexicon
- A new priming paradigm for metrical structure
- Segmental overlap between prime and target boosts the size of the priming effect
- The segmental boost occurs only in heads, similar to the lexical boost

A ‘segmental boost’ in production priming of stress

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ABSTRACT

Abstract, or supra-segmental, representations of word stress are taken as a given in much of phonology, acquisition, and psycholinguistics. However, in contrast to other abstract linguistic structure, priming effects based on stress are rarely found. Rather, priming effects are found in lexical decision times only when there is substantial segmental overlap between the prime and the target, leading researchers to hypothesize that stress is stored not abstractly, but as part of segmental representations. This study tests a possible explanation for this contradiction: that stress priming, like syntactic structural priming, can be observed readily in structure choice, but in reaction times only when internal material overlaps between prime and target. Participants’ structural choices for stress were obtained by asking them to interpret ambiguous orthographic forms compatible with multiple stress patterns (e.g. RECORD). These interpretations were influenced by the stress pattern of a preceding prime, even when no segments overlapped. When segments did overlap, the size of the priming effect increased, but only when segments overlapped in the stressed syllable - a head position. The paper concludes that stress is abstractly represented in lexical items, and suggests that metrical priming and syntactic priming may share an underlying mechanism.

1. Introduction

Language use is governed at many different levels by abstract structure that is not readily observable, and must be inferred by analysts from distributions of observable forms. However, in many cases the same observable forms are compatible with many hypotheses regarding the nature of the abstract structures which might give rise to them. This paper looks specifically at word stress, a phenomenon which has been hypothesized to be governed by abstract structures such as metrical frames, grids, or metrical feet. In the domain of theoretical phonology, abstract stress structures explain generalizations within languages and in the typology of stress patterns across languages. However, in the domain of speech processing, priming results have been used to argue against any abstract representation of stress at all, at least in the lexicon. In this paper, I will argue that this apparent contradiction in the literature can be resolved by comparing metrical structure to syntactic structure. Like syntactic structure, metrical structure exhibits priming more robustly when a choice of structure is being made than when ease of processing alone is measured.

Using a new paradigm involving the production of real words of English, in which speakers interpret visual stimuli which are ambiguous in their stress pattern (e.g. RECORD), I demonstrate that metrical structure does induce priming in real words, and furthermore that metrical priming is like syntactic priming in another way: segmental overlap from prime to target increases the size of the priming effect, but only when that segmental material is positioned in the structural head - the stressed syllable. This finding parallels findings about the lexical boost in syntactic priming, an effect whereby lexical overlap between a prime structure and target structure increases the size of the priming effect, but only if the overlapping material is the syntactic head of the structure being primed.

1.1. Is stress abstractly represented?

In languages, like English, in which a word’s stress pattern is not completely predictable, lexical entries must contain information about each word’s stress pattern. For example, speakers must know that PERmit (noun) has stress on the first syllable, while perMIT (verb) has stress on the second syllable. Indeed, studies of lexical access do consistently demonstrate that stress information is available to speakers during the lexical access process. In Spanish, both cross-modal priming (Soto-Faraco, Sebastián-Gallés and Cutler, 2001) and visually-presented masked priming (Gutiérrez-Palma and Palma-Reyes, 2007) demonstrate that listeners recognize words more slowly and with more errors when they are preceded by a prime that mismatches in stress than when they are preceded by a prime that matches in stress. For example, Soto-Faraco et al. used auditory fragments as primes for target words like PRINcipe (initial stress), which either matched in stress (fragment PRINci-) or mismatched (fragment prinCI- from the word prinCIpio). In English (Cooper, Cutler and Wales, 2002) and in Dutch (Donselaar, Kostner and Cutler, 2005) both bisyllabic (ADmi- from ADmiral and admi- from admiRAtion) and monosyllabic (mus- from MUsic/muSEum) primes facilitate lexical access of a subsequent target that matches in stress. Looking at Italian, Tabossi and Tagliapietra (2005) took this methodology a step further, finding that stress match produced facilitation in lexical decision, even when the stress matched a semantic neighbor of the target and not the actual target (example: the prime [ˈgo.mi] matching gomito ‘elbow’ where the target was target braccio ‘arm’).

In the above studies, primes and targets always matched in segmental material. This fact turns out to be crucial:

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Schiller, Fikkert and Levelt (2004) examined Dutch speakers' picture-naming reaction times when pictures were precedes by primes that either matched or mismatched in stress, but did not share any segmental material. For example, a picture of a trout (*foREL*) would be preceded by the prime *vamPIER* ('vampire', matching) or by *KEtel* ('kettle', mismatching). Over the course of three experiments, they found no effect of stress match, even though there were significant effects of semantic relatedness. Protopapas, Panagaki, Andrikopoulou, Gutiérrez Palma and Arvaniti (2016) also tested the effect of segmental overlap in Greek: in four experiments, stress match alone did not affect lexical decision reaction times, but in a fifth experiment, with segmental match between prime and target, stress match again did affect lexical decision times. In contrast to both of these studies, however, Domínguez Martínez and Cuetos Vega (2018) found facilitation in a Spanish lexical decision task for pairs like PERsa/RASgo, which matched in stress but not in segmental content, compared to unrelated pairs (dorMI/RASgo). This study used a very short SOA (32ms), and included orthographic marking of stress where it was appropriate (e.g. dormí).

Protopapas et al. conclude based on their detailed study that segmental match is required for stress priming because stress is in fact not represented separately from segments. That is, it is not truly 'suprasegmental'. In languages with vowel reduction, like English, stress patterns could be hard-coded in the lexicon via vowel quality. For example, the words REcord and reCORD in English are pronounced with different sequences of vowels: [ɹə.ˈkɔɹd] vs. [ˈɹɛ.kɻd], and could in principle be stored in the lexicon this way too, as [ɹəkɔɹd] and [ɹɛkɹd]. However, not all vowels reduce, even in a language with robust vowel reduction like English. Consider the example above of PERmit vs. perMIT. In both words, the first vowel is a syllabic [ɹ], while the second vowel is [ɪ]. For some speakers the [ɪ] may be reduced to [ɨ], a vowel that is still phonetically very similar to [ɪ]. One of the reasons Protopapas et al. studied Greek is that Greek has very little vowel reduction, and so words' lexical entries cannot be distinguished only by vowel quality. Rather, they argue, phonetic differences between stressed and unstressed syllables, such as duration, pitch, intensity, or spectral tilt, must be stored within the segments of a lexical item. For them, the [i] of *FImosi* would differ from the [i] of *fiLAme* in its lexical representation: The stressed [i] would include specifications for long duration, high intensity, and so on, while the unstressed [i] would include specifications for short duration, low intensity, etc.

I will refer to the hypothesis that stress is stored as phonetic information on a particular segment as the Local Representation Hypothesis, following Moore-Cantwell, Bosch, Kahn, Kim and Shoemaker (2024). While the Local Representation Hypothesis does explain the lexical decision results, it stands in contrast to theories of stress representation more broadly (e.g. Hayes, 1995; Gordon, 2002; van Oostendorp, 2015) which posit representations of stress that are suprasegmental and do not interact directly with segments. In fact, stress patterns are generally thought of as assigning structure to syllables, rather than directly to the segments of a word.

This suprasegmental representation of stress is supported phonetically by the fact that the phonetics of stress is not consistent across utterances, even for a given word. In many languages, stressed syllables host pitch accents, which can have a variety of actual pitch contours determined by syntactic, semantic, and pragmatic context (Pierrehumbert, 1980; Ladd, 2008). Other phonetic cues to stress are malleable as well, subject to rapid adaptation and recalibration (Bosker, 2022; Bujok, Peeters, Meyer and Bosker, 2025b). Listeners can even incorporate atypical cues for stress, such as beat gestures, into their perceptual system (Bujok, Meyer and Bosker, 2025a).

The categorization of syllables into "stressed" and "unstressed" can affect choices of higher level structure as well, the most obvious case being the alignment of stressed syllables with strong positions in poetic meters (Jakobson, 1960; Morris and Keyser, 1971; Blumenfeld, 2016). For metrical alignment, one stressed syllable is as good as another, and in fact the whole point of writing in stress-based poetic meter is to create parallelism from line to line in the patterning of stresses. Stress can also influence both word choice and constituent ordering at the syntactic level (Hilpert, 2008; Anttila, Matthew and Michael, 2010; Shih, 2017; Breiss and Hayes, 2020), indicating that it is probably part of our online representations during production.

Abstract stress representations are also required in order to explain the regularity in stress patterns, even in languages with lexically-specified stress such as English, Spanish, or Greek. These regularities are often productive, and can be applied by speakers to novel words, regardless of specific segmental content (Fidelholtz, 1975; Guion, Clark, Harada and Wayland, 2003; Domahs, Plag and Carroll, 2014; Garcia, 2017; Moore-Cantwell, 2020). Even very early in acquisition, children as young as 9 months old exhibit awareness of these stress regularities (Jusczyk, Cutler and Redanz, 1993), and can use them to help segment speech into words (Thiessen and Saffran, 2003; Kabak, Maniwa and Kazanina, 2010).

Across languages, stress patterns adhere to certain typological generalizations (Goedemans and van der Hulst, 2009). For example, stress tends to be alternating: languages disprefer multiple stresses in a row (as in bàndána), or sequences of unstressed syllables (cátamaràn). Stress also tends to be edge-aligned. Languages exhibit patterns like "stress the first syllable", "stress the last syllable", or "stress the penultimate syllable", but never "stress the middle syllable". Thus, researchers propose not only that stress is represented abstractly, but that it is represented in specific ways that help to explain these typological generalizations Hayes (1995); Gordon (2002); van Oostendorp (2015).

One aspect of the typology of stress is particularly relevant to the plausibility of the Local Representation Hypothesis: It is not just that stress generalizations need not be tied to any specific segmental content, but rather that they

seemingly *cannot* be. Patterns in which syllables beginning with certain consonants attract stress, or in which certain vowel qualities attract stress, seem to be nearly unattested, even though an extensive literature exists examining putative evidence for them (see Shih and De Lacy, 2019 for an overview). When segmental content does influence stress placement, it does so via the abstraction of syllable weight. Disparate segmental factors like coda consonants, vowel length and quality, and even number of onset consonants, can influence whether a syllable is classified as "light" or "heavy" in a language, and it is that classification in turn that affects stress placement Ryan (2019).

While the above points constitute evidence that stress is represented suprasegmentally, abstracted away from segments, there is much less clarity as to what the nature of that suprasegmental representation might be. Metrical feet are a commonly used representational device, but stand in contrast to grid-based approaches like Hayes (1983). Within both approaches, what sorts of representations are possible, and what appear in specific languages, is the subject of a great deal of research. Some examples of open questions in the metrical feet literature include: whether metrical feet can be recursive or not (Bennett, 2013; Martínez-Paricio and Kager, 2016), whether syllables that are unstressed can ever constitute a 'degenerate' foot (Ryan, 2022), and whether left-alignment and right-alignment of feet are both attested (Alber, 2005; Asherov, 2023). Another issue of great interest, and which will be discussed in some detail below, is whether or not stress is stored as such in the lexicon, and if so whether it makes use of the same representations that give rise to stress generalizations within and across languages.

Research on representational questions such as these could greatly benefit from psycholinguistic study, and in particular the use of a priming paradigm to probe structural details, in much the same way that the study of syntactic structure has benefitted (Pickering and Ferreira, 2008; Cai and Zhao, 2024). However, there is an apparent contradiction in the literature: While abstract stress representations seem to be necessary to explain both the phonetics of stress, and its patterning within and across languages, those same representations do not appear consistently in priming studies. A resolution to this contradiction has the potential to increase our understanding of stress representations themselves, as well as the mechanism(s) behind priming effects.

1.2. Structure choice, not timing measures

One proposal for such a resolution comes from Moore-Cantwell et al. (2024), who demonstrate priming of abstract stress patterns in the production of nonwords. They argue that priming of stress patterns is analogous not to priming of segments, but to priming of syntactic structure, which is also abstract. Priming effects for syntactic structure are often not observed in timing measures such as latency or reading times unless there is lexical overlap between prime and target (Pickering and Ferreira, 2008; Segaert, Wheeldon and Hagoort, 2016; Tooley, 2023). When participants are making choices about structure, however, priming is observed without this overlap. Moore-Cantwell et al. argue that stress behaves like

> priming of abstract stress patterns has generally not been observed in lexical decision times precisely because this is a timing measure. They hypothesize that priming is observed in lexical decision times when segments overlap for the same reason priming is observed in syntactic structural priming when words overlap: segmental overlap "boosts" the priming effect. In order to see priming effects for abstract stress when no segments overlap, they measure participants' choices of which stress pattern to assign to a nonword, and find that those choices were affected by the immediately preceding stress pattern regardless of any segmental overlap.

Indeed, other studies have found some evidence for priming of abstract stress patterns by using measurements other than lexical decision times. Many have used event-related potentials (ERP) to find evidence for facilitated processing of words after a stress-matched prime. Friedrich, Kotz, Friederici and Alter (2004) found a reduced amplitude of the P350 for processing of a target preceded by a stress match compared to one preceded by a stress mismatch. Schild, Becker and Friedrich (2014) found no effects of stress match on reaction times, but did see differences between stress match and mismatch conditions in ERPs. Sulpizio, Arduino, Paizi and Burani (2013); Sulpizio, Vespignani and Job (2016) found similar effects in Italian.

A few studies have also tested priming of stress in production by measuring how many stress errors participants produce (FORget instead of forGET, for example). Colombo and Zevin (2009), looking at Italian, found more incorrect stresses after a nonword with dominant stress than after a real word with non-dominant stress. Shaw (2012, 2013), in English, used a blocked design and a speeded production task, and found that participants produced more errors like FORget after initial-stressed primes. Final-stressed primes did not appear to have the same effect on production however.

1.3. Properties of the lexical boost

The 'lexical boost' occurs in structural priming when the same lexical items occur in both the prime and the target, causing the priming effect to be larger than it is when no lexical items overlap (Pickering and Branigan, 1998). The effect has been attributed to the presence of residual activation on lemmas, structures, and crucially their connections. This residual activation leads to a priming effect stronger than that of the structure or the lemma alone. If Moore-Cantwell et al. (2024) are correct, then a similar effect occurs with repeated segments within a metrical structure, so that priming effects for stress are much larger when segments overlap than when they do not. If this parallel holds, we might expect segmental overlap in stress priming to behave similarly to the lexical boost effect.

In particular, the lexical boost typically only occurs when the repeated lexical item is the head of the structure being primed. While a few studies have shown a boost effect for

repeated lexical items which are not heads in experiments (Scheepers, Raffray and Myachykov, 2017; McLean, Pickering and Branigan, 2004), and in corpora (Reitter, Keller and Moore, 2011; Snider, 2009), these studies typically involve the repetition of several lexical items in a row. When the boost effect is localized to a single lexical item, it seems to be limited to syntactic heads. Cleland and Pickering (2003) find priming of noun phrase structure, comparing structures like *the red square* to *the square that's red*. Here, a repeated noun caused a boost in the priming effect, but a repeated adjective did not. Similarly, Carminati, van Gompel and Wakeford (2019) found a boost for repeated verbs in priming prepositional vs. direct object structures within a ditransitive verb. Only repeated verbs caused a boost, not repeated subjects, indirect objects, or direct objects (see also Huang, Liu, Lu, Sun, Wang, Branigan and Pickering, 2023; van Gompel, Hartsuiker and Kantola, 2026). Kantola, van Gompel and Wakeford (2023) compare repetition of the ditransitive verb to repetition of a higher matrix verb in structures like *The hotel owner decided to loan the tourist a tent/a tent to the tourist*. Here, repetition of *loan* caused a boost but repetition of *decide* did not.

That heads induce a lexical boost more than nonhead lexical items supports models of priming which rely on residual activation, rather than those based on implicit learning (such as Chang, Dell and Bock, 2006). This understanding of the lexical boost relies on a model of the lexicon in which lemmas can be linked to nodes containing combinatorial information (Pickering and Branigan, 1998). For verbs, this combinatorial information is somewhat similar to subcategorization information: specific verbs have specific requirements for what kind of structures they can take as complement. Many verbs allow multiple kinds: when a complement structure and a verb are both activated, the connection between them is activated as well, and can boost the likelihood that a speaker will use that verb in that structure again.

In terms of stress, words also have a head: it is the stressed syllable (Liberman and Prince, 1977). Should we expect segmental repetition to boost a stress priming effect more when it is part of the stressed syllable than when it is part of an unstressed syllable? In principle, there could be residual activation on segments, a stress pattern, and their connection, which would affect subsequent stress production and comprehension in much the same way as residual activation on a repeated verb and its associated structure affects sentence production. However, neither segments nor syllables have the same relationship to stress patterns that lexical items do to syntactic structure. Segments and syllables do not subcategorize for particular stress patterns, nor do they have an equivalent of a syntactic category. For these reasons, we might expect segmental overlap anywhere in a word to boost stress priming equally well.

Moore-Cantwell et al. (2024) provide some preliminary evidence regarding segmental overlap: They find an effect of overlapping stressed syllable nucleus, so that after a nonword like [ˈbæməki], a subsequent nonword sharing the [æ] nucleus was even more likely to take initial stress ([θæləsi]) than a nonword with a different nucleus ([fɑməkə]/ [fəmɑkə]). They also find an effect of overlapping final vowels, which for them were never stressed. For example, [ˈbæməki] would be more likely to be followed by initial stress in a nonword that also ended in [i]. This finding suggests that the location of segmental overlap does not matter, and unlike the lexical boost, a segmental boost would not rely on repetition of head material specifically. On the other hand, Protopapas et al. (2016) used items which all had the same initial sequence ('fi'), and yet they found no priming at all in their lexical decision tasks for prime-target pairs like fiSA ∼ fiLAme. They only found priming when segmental material overlapped completely, as in fiLA ∼ fiLAme. If segmental overlap anywhere in the word was sufficient to boost the priming of stress to the point where it can be detected in reaction times, then we would have expected some priming effects for pairs like fiSA ∼ fiLAme.

So, the evidence from current literature on whether headedness matters for segmental overlap in stress priming is mixed. However, neither of these studies were designed to directly test this issue, and did not control for it. In Moore-Cantwell et al.'s study, only a small percentage of items matched in stressed vowel, and only two final vowels were used in the study, possibly making final vowel salient to participants, and subject to explicit memory effects. Protopapas et al. used only items which matched in the first syllable, they did not test these against any items which matched in other syllables but not the first, nor items which did not overlap at all in segmental material.

1.4. Stress in the lexicon

Although there is a great deal of evidence for the abstract representation of stress in grammar and in production, it is much less well understood whether stress is represented similarly in existing lexical items as it is in phonological generalizations. In theoretical phonological models of stress, it is either assumed to not be stored in the lexicon, or to be stored only when it is exceptional, diverging from the default of the language (e.g. Chomsky and Halle, 1968). Broś, Meyer, Kliesch and Dellwo (2022) use ERPs to argue that regular (default) stresses are not listed in the lexicon, but exceptional stresses are. Similarly, Peperkamp, Vendelin and Dupoux (2010) show that speakers' ability to encode stress in their lexicon at all is dependent on how regular stress is in their language (how many exceptional words exist). On the other hand, Andrikopoulou, Protopapaps and Arvaniti (2021); Garcia and Guzzo (2022) fail to find evidence, in Greek and Portuguese respectively, that stress is stored any differently on exceptional and non-exceptional words.

While Colombo and Zevin (2009); Shaw (2013); Moore-Cantwell et al. (2024) all found evidence of stress priming in production, in all cases their results can be interpreted as evidence for the existence of abstract stress representations during production, but not that those representations are

actually embedded in lexical entries. Moore-Cantwell et al. used only nonwords, and Colombo and Zevin and Shaw examined stress errors - specifically, productions that would not align with a lexical item's encoded stress pattern. It remains unclear then, whether lexical decision tasks fail to find priming of abstract stress patterns because lexical decision is a timing measure, or because stress is represented in the lexicon according to the Local Representation Hypothesis, even though abstract stress representations can be used in production.

Furthermore, stress priming in lexical decision tasks when segments overlap is not necessarily evidence of any particular kind of stress representation in lexical entries: Previous studies of stress priming with overlapping segments compare only fully overlapped word fragments (such as ADmi ~ ADmiral). As Protopapas et al. (2016) point out, priming obtained from overlapping structures like this could reflect identity priming, rather than any kind of effect of specific segments or specific metrical structure.

1.5. The present study

Experiment 1 seeks to establish the existence of abstract stress representation in the lexicon, by eliciting correct productions of real words. In line with cases of syntactic priming in comprehension (Tooley, 2023), written forms are used which can be interpreted as two different words, with different stress patterns (e.g. DESERT). Participants interpret the written form as one word or the other, so that a priming effect can be measured in participants' choice of interpretation.

If priming is observed in real words, this would constitute evidence against the Local Representation Hypothesis, and for what I will call the Segmental Boost Hypothesis. The weak version of this hypothesis is stated in (1):

(1) **The Segmental Boost Hypothesis** (weak version): Stress is represented abstractly in the lexicon. This abstract representation, like representations of syntactic structure, does not induce priming effects in timing measures unless 'boosted' by overlap of segmental material between prime and target, but can induce priming in outcome measures.

A stronger version of the Segmental Boost Hypothesis would draws a tighter connection between syntactic priming and stress priming:

(2) **The Segmental Boost Hypothesis** (strong version) Priming effects for abstract metrical structure can be boosted by segmental overlap of structural heads (stressed syllables), but not by other segmental overlap.

If this stronger version of the hypothesis is true, then this would imply that the segmental boost is reflective of the same underlying mechanism as the lexical boost, and therefore models of the lexical boost can and should be accountable for the behavior of the segmental boost as well. Although this very strong position cannot be justified effectively in a single paper, Experiment 2 will at least test the strong version of the Segmental Boost Hypothesis.

Experiment 2 uses a similar methodology to Experiment 1, but manipulates segmental overlap directly, comparing overlap in words' stressed syllables to overlap in unstressed syllables, and comparing that to no overlap anywhere in the word. If overlap in stressed syllables boosts the size of the priming effect, but overlap in unstressed syllables does not, this would support the strong version of the Segmental Boost Hypothesis. If both kinds of overlap boost the size of the priming effect (even if stressed syllable overlap boosts more), this would support only the weak version of the Segmental Boost Hypothesis, suggesting similar dyanmics between metrical priming and syntactic priming, but that they are likely due to different mechanisms. If no effect of segmental overlap is found at all, this would call into question even the weak version of the Segmental Boost Hypothesis. It would suggest that abstract stress representations do exist in the lexicon, but that the explanation given by this hypothesis for the behavior of stress in lexical decision tasks is flawed.

2. Experiment 1

Experiment 1 tests whether recent experience with abstract stress representations can affect the processing of real words. The paradigm uses homograph pairs in English, such as DESERT, which when read can be interpreted as DEsert, a noun, or as deSERT, a verb. Words like this are presented visually, preceded by either a stress matched or stress mismatched prime, neither of which overlaps in segmental material with either pronunciation of the target. Participants produce (read aloud) both the prime and the target, and subsequently use the target in a sentence, illustrating which word they interpreted it to be. If stress representations persist from prime to target, participants will be more likely to interpret the written word as the homograph which matches the primes stress pattern. After LIMBO, an initial-stressed word, we would expect more interpretations of DESERT as the arid ecosystem, whereas after ACCLAIM, a final-stressed word, we would expect more interpretations of DESERT as an act of leaving behind.

2.1. Methods

2.1.1. Items

A total of 40 homograph pairs were selected in which one orthographic form corresponded to two words with different stress patterns, and different parts of speech. The majority of these were semantically related noun-verb pairs, such as [rɛkɹd] (N) and [rəkɔɹd] (V), both spelled RECORD. However, some were not semantically related (DESERT) and some were noun-adjective pairs (COMPLEX). Lemma frequencies were collected from CELEX (Baayen, Piepenbrock and van H, 1993) for both words in the pair. initial-stressed words had log-frequencies ranging from 0.95 to 3.27 (μ=2.16), and final-stressed words had log-frequencies ranging from 0.95 to 3.25 (μ=2.23). Additionally, pairs were

selected so that for half of them the initial-stressed word was more frequent (average difference = 0.39) and for half the final-stressed word was more frequent (average difference = 0.53). Primes were selected for each pair, in which the stress pattern was unambiguous from the orthography. One initial-stressed prime and one final-stressed prime was selected, neither of which overlapped in segmental content with the members of the word pair. Frequencies were obtained for primes using $\text{SUBTLEX}_{\text{US}}$ (Brysbaert and New, 2009)[1]. Primes had a maximum log frequency of 2.98 and a minimum of 0.78, with a mean of 1.83.

Items were counterbalanced across participants and trials so that each participant saw each target exactly once, and saw an equal number of initial-stressed primes and final-stressed primes (20 each). Across participants, each target appeared with each prime an equal number of times.

2.1.2. Participants

Participants were UCLA undergraduates, recruited through UCLA's SONA system. They received course credit for linguistics or psychology courses by participating. All were native speakers of English, and over 18 years old. A total of 74 participants completed the study.

2.1.3. Procedure

Participants completed the experiment online, via the PCIbex platform (Zehr and Schwarz, 2018). After completing the consent form and a sound test, participants were instructed to read each word aloud, and when prompted use the word they had just read in a sentence, which they then typed into a text box. Participants read aloud both prime and target, and were not made aware of the prime-target structure of the study, though they likely would have noticed that they were asked to produce sentences for every other word.

Words were presented in all caps, in 38pt font (50 pixels in HTML). Participants were recorded as they read the words aloud, though these recordings will not be analyzed here.

2.1.4. Analysis

Participants' written responses were inspected by hand, and labeled as *initial stress* or *final stress* based on the meaning and part of speech apparently intended by the sentence. This labeling was carried out either by the author or by a trained research assistant. All labels were checked by a second transcriber, and there were very few disagreements (fewer than 1% of labels). Disagreements were either adjudicated by a third transcriber or, where appropriate, marked as an error. Responses were marked as an error if they did not contain the target word, did not clearly show its part of speech, or if they used the target word in both available senses.

A Bayesian logistic regression was conducted in R(core team, 2025) using the brms package in R (Bürkner, 2017). The dependent variable was target stress, as inferred from the participant's use of the target in a sentence. This was coded as 0 for initial stress, and 1 for final stress. Predictors were the lexical stress of the prime (0 for initial, 1 for final), the prime's log frequency, and the target's continuous bias for initial stress, calculated as the log frequency in CELEX of the initial-stressed homograph minus the log frequency of the final-stressed homograph. Recall that half of the targets were biased toward initial stress, and thus had a positive value for this measure, and half were biased toward final stress, and thus had a negative value. Values ranged from -1 to 1.

All three factors and their interactions were fitted, and a full random effects structure for participants and for items was fit as well. Slopes for prime frequency and target bias could not be fit because neither varied sufficiently within target. Target bias is a property of the target itself, and therefore varies not at all, and each target had just two assigned primes, so the prime frequency also had only two possible values for each target. The final model equation is given in (3).

(3) Model structure

```
target stress ~ Prime stress * Target bias * prime
freq. + (1+ Prime stress * Target bias * prime
freq.| Participant) + (1+ Prime stress | Target)
```

Weakly informative priors (Normal(0,10)) were used for all coefficients, and a Student's t(3,0,2.5) prior was used for the intercept and all standard deviations. Additionally, an uninformative (lkj, eta=1) prior was used on the correlations between random effects. Bayes factors were estimated using the Savage-Dickey density ratio method, via the `hypothesis()` function in `brms`. The model was fit using 4 chains of 20,000 iterations each, with a warmup period of 2000 iterations.[2]

Stress	POS	Response
Initial	N	*I played on my video game* **console** *everyday.*
Initial	N	*Aim your remote at the* **console**
Initial	N	*Pick up the* **console**
Final	V	*I* **consoled** *my friend after he got a bad grade.*
Final	V	*I* **consoled** *my friend going through a breakup*
Initial	N	*Don't lose the* **console** *for this game.*
Final	V	*The mother* **consoled** *her daughter.*
Final	V	*i must* **console** *her.*
Initial	N	*my brother's gaming* **console**
Final	V	*I* **console** *others.*

Table 1
Example written responses

2.2. Results

In general, participants provided coherent responses that clearly demonstrated which word out of a given homograph pair they had in mind. Some example responses after the target CONSOLE are given in 1.

[1] Note that $\text{SUBTLEX}_{\text{US}}$ does not contain lemma frequencies, but rather orthographic frequencies. This is sufficient to estimate prime frequencies, but could not be used to distinguish the frequencies of each homograph.

[2] All data and analysis scripts for this paper are available at https://osf.io/y32qw/

	Estimate	95% CI		Bayes factor	
				β= Estimate	β=0
Intercept	-0.30	-1.56	0.94		
Prime stress (F)	1.25	0.09	2.45	16.75	1.87
Prime freq.	0.18	-0.37	0.74	36.28	28.63
Target Bias	-2.28	-5.00	0.42	7.45	1.85
Prime Stress x Target Bias	1.87	-0.41	4.16	8.91	2.30
Prime Stress x Prime freq.	-0.45	-1.07	0.15	32.22	10.87
Target Bias x Prime freq.	0.51	-0.67	1.71	16.82	11.67
Prime Stress xTarget Bias x Prime freq.	-0.69	-1.85	0.46	17.32	8.40

Table 2
Fitted values for a Bayesian mixed-effects regression, with Bayes factors reported.

	Target stress interpretation	
Prime stress	Initial	Final
Initial	710	708
Final	640	783

Table 3
Counts of response types after initial vs. final stressed primes

Fitted estimates, 95% confidence intervals, and Bayes factors are reported in 2. Positive coefficients indicate a greater probability of final stress interpretations of the target, while negative coefficients indicate a greater rate of initial stress. For example, Target Bias is associated with a greater rate of initial stress interpretations because items with higher values on this variable are more biased toward the initial stress interpretation. The only factor with a confidence interval excluding 0 was Prime stress. This coefficient is positive, indicating that primes with final stress were more likely to be followed by a final stress interpretation of the target. The effect of prime stress on target stress interpretation is shown numerically in 3.

Bayes factors are reported for each effect in the fitted model, indicating the strength of evidence for each. Two Bayes factors are reported: the strength of evidence for the hypothesis that the effect is equal to the model's estimate, and the strength of evidence for the hypothesis that the effect is zero. A Bayes factor greater than 10 indicates strong evidence (Lee and Wagenmakers, 2014), 3-10 indicates moderate evidence. In this case, some factors, such as `Prime Freq.` have strong evidence for their actual estimated value, and strong evidence that their value is zero. The effect of Prime Stress has a Bayes factor of 16.75 for its estimate of 1.25, and very weak or no evidence (Bayes factor of 1.87) that its effect is zero.

2.3. Discussion

Experiment 1 found a priming effect for stress evident in participants' interpretations of stress-ambiguous orthographic forms. This result provides evidence for the abstract representation of stress in lexical entries, and aligns with syntactic priming results in which priming effects appear in structural choice, rather than in timing measures. Here, participants make a choice about how to interpret a written form with no context; it appears that a recently encountered stress pattern can push participants toward the interpretation that matches that stress pattern.

Unlike previous production studies that found stress priming (Colombo and Zevin, 2009; Shaw, 2012, 2013; Moore-Cantwell et al., 2024), Experiment 1 finds priming of existing listed stress patterns. Primes affect which lexical entry is accessed when participants see a certain orthographic form, indicating that an abstract stress pattern can be used in the lexical access process, in a way that matches some lexical entries and does not match others. This is evidence that abstract stress representations are part of lexical entries.

3. Experiment 2

In Experiment 1, abstract stress representations persisted from a prime to a stress-ambiguous orthographic target. Participants chose from two possible interpretations of that target, as initial stress or final stress. Experiment 2 replicates Experiment 1, and also includes primes which overlap in segmental material, either in the stressed syllable only, or in the unstressed syllable only, in order to directly test the Segmental Boost Hypothesis. A stressed syllable overlap can be initial (**DE**mo ~ **DE**sert) or final (ex**ERT** ~ de**SERT**). Likewise, an unstressed syllable overlap can be initial (**de**FLATE ~ **de**SERT) or final (SWEAT**shirt** ~ DE**sert**).

A larger priming effect when segments overlap than when they do not (as in LIMbo ~ DESERT, or accLAIM ~ DESERT) would support both versions of the Segmental Boost Hypothesis. This would provide evidence that stress is represented abstractly in lexical items, and explain why stress priming appears in lexical decision only when segments overlap. If the priming effect is boosted only when *stressed* segments overlap, this would constitute evidence that segmental overlap acts in metrical priming very similarly to how lexical overlap acts in syntactic priming, in that overlap in head positions specifically, and not nonhead positions, causes a boost. This finding would support the strong version of the Segmental Boost Hypothesis.

3.1. Methods

3.1.1. Items

From the 40 homograph pairs used in Experiment 1, 24 were selected for which a complete set of primes could be found. Each target was assigned six possible primes, which

Target: **DESERT**

Stress	Prime	Overlap	Transcription	Match
Initial	DEMO	stressed	ˈ**d**ɛ.mo͡w	ˈ**d**ɛ.z̩t
Final	EXERT	stressed	əg.ˈ**z̩t**	də.ˈ**z̩t**
Initial	SWEATSHIRT	unstressed	ˈswɛt.ʃ̩**t**	ˈdɛ.**z̩t**
Final	DEFLATE	unstressed	**d**ə.flе͡jt	**d**ə.ˈz̩t
Initial	LIMBO	none	ˈlɪm.bo͡w	-
Final	ACCLAIM	none	ə.ˈkle͡jm	-

Table 4
Example Item

were either initially or finally stressed, and which either overlapped with the target in stressed or unstressed position, or not at all (these were the same primes used in Experiment 1). An example prime set is given in (4):

Overlapping syllables overlapped with the stressed/unstressed syllable of the version of the target with the same stress pattern as the prime. For example, in (3), the stressed syllable of DEMO, [dɛ], matches the [dɛ] of DEsert (N), not the [də] of deSERT (V). Likewise, the stressless syllable of DEFLATE, [də], matched the [də] of deSERT(V), not the [dɛ] of DEsert (N).

Overlapping syllables always shared the same nucleus, and codas and onsets were matched as closely as possible. For example, [ʃ̩t] of SWEATSHIRT matches [z̩t] in nucleus and coda. The onset does not match, but this prime was chosen over others that mismatched in onset because [ʃ] shares many features with [z]. A full list of primes for each item are given in the Appendix. Selected primes had an average log frequency (obtained, as in Experiment 1, from $\text{SUBTLEX}_{\text{US}}$ Brysbaert and New, 2009) of 1.41, with a minimum of 0.30, and a maximum of 2.00.

As in Experiment 1, the set of items was balanced according to each target's bias to take initial stress, based on the relative log frequencies of the initial-stressed homograph and the final-stressed homograph in CELEX (Baayen et al., 1993). Half of items were biased to take initial stress (mean log frequency difference of 0.40), and half were biased to take final stress (mean log frequency difference of 0.59).

Items were counterbalanced across runs of the experiment so that each participant saw each target exactly once, and targets were assigned to different conditions so that a participant saw each condition 4 times. Across participants, each target occurred a roughly equal number of times with each of its assigned primes (after exclusions of trials with errors, the ultimate number of trials per target ranged from 27 to 39).

3.1.2. Participants

Participants were UCLA undergraduates, recruited through UCLA's SONA system. They received course credit for linguistics or psychology courses by participating. All were native speakers of English, and over 18 years old. A total of 209 participants completed the study. Note that a greater number of participants were recruited for Experiment 2 compared to Experiment 1 (more than twice as many), because Experiment 2 had far fewer items, and more conditions, so each participant contributed just 4 data points per condition.

3.1.3. Procedure

The procedure was identical to that of Experiment 1.

3.1.4. Analysis

Analysis proceeded similarly to that of Experiment 1. As in Experiment 1, labeling disagreements for participants' generated sentences were rare - less than 1%.

As in Experiment 1, a Bayesian logistic regression was conducted using `brms`. The dependent variable was target stress, inferred from the participant's use of the target in a sentence as before, and coded as 0 for initial stress, and 1 for final stress as before. Predictors were the lexical stress of the prime (0 for initial, 1 for final), and the type of segmental overlap between prime and target. This factor, OVERLAP was Helmert-coded so that the *no overlap* condition could be compared to the combination of *unstressed overlap* and *stressed overlap*, and *stressed overlap* could be directly compared to *unstressed overlap*. Because neither prime frequency nor target bias affected the model outcomes in Experiment 1, they were not included here.

A full random effects structure was fit as well, with prime stress, overlap and their interaction by both participants and by targets. Note that while prime frequency and target bias were not included, random intercepts and slopes for targets will fit variance introduced by target-specific properties like bias and the frequency of a target's assigned primes.

(4) Model structure

```
target stress ~ Prime stress * Overlap + (1+ Prime
stress * OvePrime stress * Overlap | Participant)
+ (1+ Prime stress * Overlap | Target)
```

As in Experiment 1, weakly informative priors (Normal(0,10)) were used for all coefficients, a student's t(3,0,2.5) was used for the intercept and standard deviations, and an lkj prior (eta=1) was used for the correlations between random effects. As before, the model was fit using 4 chains of 20,000 iterations with a 2000 iteration warmup period.

3.2. Results

Fitted estimates, 95% confidence intervals, and Bayes factors are reported in Table 5. As in Experiment 1, positive coefficients indicate a greater probability of final stress, and negative coefficients indicate a greater probability of initial

	Estimate	95% CI		Bayes Factor	
				β= Estimate	β=0
Intercept	0.00	-0.79	0.77		
Prime stress (F)	0.30	0.12	0.47	116.75	0.53
Overlap 1 (none vs. str + unstr)	0.04	-0.31	0.38	58.50	-
Overlap 2 (str vs. unstr)	-0.22	-0.50	0.06	71.60	20.95
Prime Stress x Overlap 1	-0.07	-0.70	0.58	33.03	-
Prime Stress x Overlap 2	0.72	0.33	1.11	51.66	0.07

Table 5
Fitted values for a Bayesian mixed-effects regression, with Bayes factors reported.

stress. For effects with an estimate very close to 0 (within 0.1), no Bayes factor is reported for β=0 since evidence that a coefficient is zero and evidence that it is slightly off from zero will be very similar.

As in Experiment 1, Prime stress has a confidence interval excluding zero, and a positive coefficient, indicating that when the prime had final stress, the target was more likely to have final stress as well. Estimates were near zero for both the main effect of Overlap 1 and its interaction with Prime stress. Recall that Overlap as a factor was coded so that Overlap 1 compares the no-overlap condition to the both the unstressed overlap and the stressed overlap condition, pooled. So, no effect was found of segmental overlap alone. There was, however, a strong interaction between Prime stress and Overlap 2, which compares the unstressed-overlap condition to the stressed-overlap condition. This interaction has a positive estimate with a 95% confidence interval excluding zero, indicating that when primes have final stress *and* have segmental overlap with the target in the stressed syllable, they are especially likely to take final stress. Finally, the main effect of Overlap 2 has a negative coefficient, indicating that when the prime's stress is initial, the baseline value for the Prime stress factor, the target's stress is more likely to be initial also. This effect has a confidence interval including zero, and while its Bayes factor for the fitted negative estimate is high (71.60, indicating very strong evidence, Lee and Wagenmakers, 2014), the Bayes factor for the hypothesis that the effect is 0 is also high (20.95, strong evidence).

The effect of Prime stress has a Bayes factor of 116.75, indicating extreme evidence for the effect, and a Bayes factor of 0.53 for the hypothesis that it is zero, indicating weak evidence against that hypothesis. The interaction between Prime stress and Overlap 2 also has a large Bayes factor (51.66, indicating very strong evidence), and a Bayes factor of just 0.07 for the hypothesis that that effect is 0, strong evidence against that hypothesis.

The effect of the interaction between Prime stress and Overlap 2 is actually slightly more than twice as large as the effect of Prime stress itself. The full posterior distributions for the main effect of Prime stress and for both interactions with Overlap are shown in Figure 1.

The posterior distribution for Prime stress x Overlap 1 centers almost exactly around zero. This visualization, along with the Bayes factor of 33.03 (strong evidence) for the hypothesis that this effect is near zero (at its estimate of -0.07), indicates not just no evidence for this interaction, but rather strong evidence that this interaction is very small or zero.

The posterior for Prime stress and for the interaction between Prime stress x Overlap 2 are both centered firmly above zero, and the plot demonstrates that the effect size of the interaction is even larger than the effect size of Prime stress itself.

3.3. Discussion

As in Experiment 1, the stress of a prime affected the stress pattern that participants interpreted the target to have. This effect was boosted when segments were repeated in the stressed syllable between prime and target, but not when they were repeated in the unstressed syllable. In fact, examination of the posterior distributions suggests not just no evidence that priming is boosted when segments overlap in unstressed syllables, but affirmative evidence that it is not.

This boost of the priming effect when stressed material is repeated but not when unstressed material is repeated lines up well with the literature on the lexical boost, which typically appears only for repeated heads, not for repeated material in nonhead positions. This experiment therefore provides evidence in favor of the strong version of the Segmental Boost Hypothesis.

This result does stand in contrast to findings in Moore-Cantwell et al. (2024), where priming was boosted when words' final vowels overlapped. In their study, these final vowels were never stressed. One possibility, raised by Dabouis and Fournier (2022), is that the final vowels used in that study actually have a special status, in that they may cue a word's membership in a particular lexical stratum. It is possibly this aspect of the final vowels which allows them to induce a boost to the priming effect, but more research is necessary.

4. General Discussion

This paper provides evidence that abstract stress representations are in fact stored in the lexicon, at least in English. This result stands in contrast to many results in the literature which use lexical decision tasks (e.g.Schiller et al., 2004; Cooper et al., 2002; Protopapas et al., 2016), and do not find priming of stress without segmental overlap

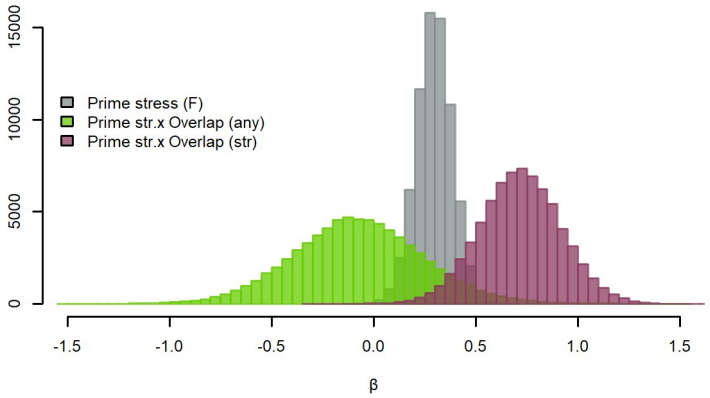


Figure 1: Posterior distributions for the effect of prime stress, and its interactions with the two types of overlap: overlap in any syllable, and overlap specifically in the stressed syllable

between prime and target. The Segmental Boost Hypothesis proposes an explanation for this contradiction: abstract stress representation is part of lexical entries, but similarly to syntactic structure, metrical structure exhibits stronger priming effects in outcome measures, where participants choose a structure in either production or interpretation, than in timing measures. The difficulty of observing priming effects in timing measures can be overcome when material is repeated between prime and target, as this repeated material induces a 'segmental boost', analogous to the lexical boost observable in syntactic priming.

Experiment 2 directly tested the Segmental Boost Hypothesis, finding that segments matching from prime to target did in fact boost the size of the metrical priming effect, but only when those matching segments occurred in stressed syllables. Since stressed syllables are the head of the word, this is evidence for the strong version of the hypothesis.

This result suggests a fairly strong parallel between metrical priming and syntactic priming not previously discussed in the literature. The two seem to exhibit similar sensitivites to experiment structure, and similar effects of repeated material. Further investigation is necessary to understand how far this parallel goes. For example, Hartsuiker, Bernolet, Schoonbaert, Speybroeck and Vanderelst (2008) show that the lexical boost decays faster than the structural priming effect itself, which can persist over relatively long periods of time. Does the same thing happen with the segmental boost in metrical structure?

If the segmental boost and the lexical boost do indeed share an underlying mechanism, an interesting issue arises: The explanation for the fact that the lexical boost appears only with repeated syntactic heads, in both residual activation and hybrid models of syntactic priming (see Tooley, 2023 for a review), relies on the special relationship of certain lexical items to the structures they occupy. Verbs, in particular, can enforce specific categorical requirements on the structure of their complements. It thus seems reasonable to imagine that, as the residual activation model claims (Pickering and Branigan, 1998), verbs have, or can have, a connection to the combinatorial nodes of the syntactic structure that other words in the structure cannot. It is activation on that connection that leads to the lexical boost effect.

While stressed syllables are thought of as the head of a word's metrical structure, they do not have the same relationship to that structure that lexical items have to the structures they head. Segments in a stressed syllable cannot, for example, determine or constrain what metrical structure other parts of the word take. Similarly, segments in a metrical structure do not have an equivalent of syntactic categories - while restrictions can exist on what segments can occur in stressed syllables, or in unstressed syllables (the obvious example being vowel reduction in unstressed syllables in many languages), these tend to be more about the range of possibilities that can occur in a given position, rather than that a certain category of segments occur in stressed position, and a different category occurs in unstressed position.

This difference between the roles of lexical items in a syntactic structure and segments in a metrical structure may indicate that the lexical boost and the segmental boost should not be modeled with the same mechanism. On the other hand, if subsequent research provides further evidence that the segmental boost and the lexical boost share common properties, this fact may help shed light on the nature of a potential shared mechanism.

Finally, this paper provides a workable paradigm with which to elicit metrical priming effects. This paradigm can be used going forward to address specific questions about the nature of metrical structure which have been difficult to address. This paradigm could be used to address the issue of whether stress is stored differently on lexical items that follow the default pattern of the language vs. lexical items which are exceptions, the question of whether stress is stored in a metrical frame (Levelt, 1993), combined with information about the word's length in syllables, or if it is stored in some more abstract way, perhaps in relation to the left or right edge of the word; or even questions around the exact structure and details of metrical feet.

A. Appendix: Items used in both experiments

	Prime		
Overlap	Initial stress	Final stress	**Target**
none	LIMBO	ACCLAIM	DESERT
str.	DEMO	EXERT	
unstr.	SWEATSHIRT	DEFLATE	I (0.34)
none	YOGA	REVOKE	INSULT
str.	INMATE	REVOLT	
unstr.	DEADBOLT	INFER	I (0.12)
none	JAZZY	ALOOF	SUSPECT
str.	SUSSEX	DISSECT	
unstr.	CROSSCHECKED	SECEDE	F (-1.07)
none	COMFY	AGLOW	INSERT
str.	INSEAM	REVERT	
unstr.	YOGURT	INFER	F (-0.90)
none	GYMNAST	ALIGHT	RECORD
str.	REFUGE	ACCORD	
unstr.	LEOPARD	RENEW	I (0.25)
none	AMMO	ALIGN	SURVEY
str.	SURPLUS	ARRAY	
unstr.	MIDDAY	SURVEIL	I (0.44)
none	BUDDHA	ALLAY	PERMIT
str.	PEARLY	OMIT	
unstr.	DEBIT	PERCEIVE	F (-0.91)
none	FLIMSY	AVAIL	EXPORT
str.	EXILE	CONTORT	
unstr.	CARPORT	EXALT	I (0.35)
none	PANDA	AMID	ESCORT
str.	ESTER	DEPORT	*bias:*
unstr.	COHORT	ESTRANGED	F(-0.04)
none	FIJI	AWRY	COMPLEX
str.	COMPOST	PERPLEX	
unstr.	VORTEX	COMPUTER	F (-0.55)
none	LUCID	AVAIL	TORMENT
str.	TORTOISE	REPENT	
unstr.	CONVENT	TOURETTE	F(-0.18)
none	GRAVY	SURPASS	CONFINE
str.	CONTOUR	REFINE	
unstr.	FELINE	CONCISE	I (0.84)
none	KAYAK	BAMBOO	PRESENT
str.	PRETZEL	ASCENT	
unstr.	FIGMENT	PREMIER	F (-0.77)
none	BANISH	DISDAIN	PROJECT
str.	PROSPER	EJECT	
unstr.	PREFECT	PROFESS	I (0.31)
none	HUMID	DIVULGE	TRANSPORT
str.	TRANSITS	COMPORT	
unstr.	SEAPORT	TRANSACT	I (0.88)
none	TALLY	BESET	COMPOUND
str.	COMPOST	IMPOUND	
unstr.	OUTBOUND	COMMUTE	F (-0.39)
none	HAGGARD	HARASS	CONVERSE
str.	CONVOY	ADVERSE	
unstr.	FEVERS	CONGEAL	F (-0.66)
none	LIVID	IMPOUND	ABSTRACT
str.	ABBESS	ENACT	
unstr.	COGNAC	ABSORB	I (0.35)
none	FUZZY	ABDUCT	PROGRESS
str.	PROCTOR	DIGRESS	
unstr.	TIGRESS	PROCURE	I (0.60)
none	TUNIC	TYCOON	ADDRESS
str.	ADMIN	DEPRESS	
unstr.	CYPRESS	ADHERE	I (0.14)
none	HYBRID	SHAMPOO	CONSOLE
str.	CONCAVE	UNROLL	
unstr.	TADPOLE	CONCEIT	F (-0.41)
none	RHUBARB	SURREAL	CONVICT
str.	CONJURE	EVICT	
unstr.	CIVIC	CONVULSE	F (-0.76)
none	PITHY	TATTOO	REBEL
str.	RELISH	MISSPELL	
unstr.	BABBLE	RESCIND	I (0.12)
none	SEWER	ERASE	CONDUCT
str.	CONQUEST	DEDUCT	
unstr.	WOODCHUCK	CONTRIVE	F (-0.39)

B. Items used only in Experiment 1

Prime			
Initial stress	Final stress	**Target**	Bias
NYLON	ANNOY	DEFECT	I (0.18)
HOLLOW	AWAIT	REFUND	I (0.41)
IRIS	AWRY	CONTENT	I (0.44)
MIGHTY	BUFFOON	DECREASE	F (-0.70)
CLINIC	CHIFFON	ADEPT	F (-0.45)
HYGIENE	DELUDE	COMPACT	F (-0.40)
FLURRY	DISDAIN	OBJECT	I (0.63)
TIMID	DISTINCT	RELAY	I (0.14)
GUPPY	EXUDE	CONTRAST	I (0.62)
ORCHARD	HARPOON	BUFFET	F (-0.44)
LILY	IMPEDE	CONTRACT	I (0.45)
ONION	JULY	EXTRACT	F (-0.30)
BURLY	MAROON	DIGEST	F (-1.0)
SCRUFFY	OBSCURE	IMPLANT	I (0.18)
SQUABBLE	SEDUCE	IMPORT	F (-0.04)
HICCUP	SUCCINCT	FORBEAR	F (-0.20)

References

Alber, B., 2005. Clash, lapse, and directionality. Natural Language & Linguistic Theory 23, 485–542. doi:0.1007/s11049-004-0482-1.

Andrikopoulou, A., Protopapaps, A., Arvaniti, A., 2021. Lexical stress representation in spoken word recognition. Journal of Experimental Psychology: Human Perception and Performance 47, 830.

Anttila, A., Matthew, A., Michael, S., 2010. The role of prosody in the English dative alternation. Language and Cognitive Processes 25, 946–981. doi:10.1080/01690960903525481.

Asherov, D., 2023. Metrical grids and active edges. Ph.D. Dissertation. Massachusetts Institute of Technology.

Baayen, R.H., Piepenbrock, R., van H, R., 1993. The CELEX lexical data base on CD-ROM. Linguistic Data Consortium.

Bennett, R., 2013. The uniqueness of metrical structure: rhythmic phonotactics in Huariapano. Phonology 30, 355–398.

Blumenfeld, L., 2016. Generative Metrics: An overview. Language and linguistics compass 10, 413–430. doi:0.1111/lnc3.12204.

Bosker, H.R., 2022. Evidence for selective adaptation and recalibration in the perception of lexical stress. Language and Speech 65, 472–490.

Breiss, C., Hayes, B., 2020. Phonological markedness effects in sentence formation. Language 96, 338–370. doi:10.1353/lan.2020.0023.

Broś, K., Meyer, M., Kliesch, M., Dellwo, V., 2022. Word stress processing integrates phonological abstraction with lexical access - an ERP study. Journal of Neurolinguistics 57, 100959.

Brysbaert, M., New, B., 2009. Moving beyond Kučera and Francis: A critical evaluation of current word frequency norms and the introduction of a new and improved word frequency measure for American English. Behavior Research Methods 41, 977–990. doi:10.3758/BRM.41.4.977.

Bujok, R., Meyer, A.S., Bosker, H.R., 2025a. Audiovisual perception of lexical stress: Beat gestures and articulatory cues. Language and Speech 61, 181–203.

Bujok, R., Peeters, D., Meyer, A.S., Bosker, H.R., 2025b. Beating stress: Evidence for recalibration of word stress perception. Attention, Perception, and Psychophysics 87, 1729–1749.

Bürkner, P.C., 2017. brms: An R package for Bayesian multilevel models using Stan. Journal of statistical software 80, 1–28. doi:10.18637/jss.v080.i01.

Cai, Z.G., Zhao, N., 2024. Structural priming: An experimental paradigm for mapping linguistic representations. Language and Linguistics Compass 18, e12507.

Carminati, M.N., van Gompel, R.P.G., Wakeford, L.J., 2019. An investigation into the lexical boost with nonhead nouns. Journal of Memory and Language 108, 104031. doi:10.1016/j.jml.2019.104031.

Chang, F., Dell, G.S., Bock, K., 2006. Becoming Syntactic. Psychological Review 113, 234–272.

Chomsky, N., Halle, M., 1968. The sound pattern of English. Harper and Row, New York, Evanston, and London.

Cleland, A.A., Pickering, M.J., 2003. The use of lexical and syntactic information in language production: Evidence from the priming of noun-phrase structure. Journal of Memory and Language 49, 214–230.

Colombo, L., Zevin, J., 2009. Stress priming in reading and the selective modulation of lexical and sub-lexical pathways. PloS one 4, e7219. doi:10.1371/journal.pone.0007219.

Cooper, N., Cutler, A., Wales, R., 2002. Constraints of lexical stress on lexical access in English: evidence from native and non-native listeners. Language and Speech 45, 207–228.

Dabouis, Q., Fournier, P., 2022. English phonologies? Models and modelisation in linguistics .

Domahs, U., Plag, I., Carroll, R., 2014. Word stress assignment in German, English and Dutch: Quantity-sensitivity and extrametricality revisited. The Journal of Comparative Germanic Linguistics 17, 59–96. doi:10.1007/s10828-014-9063-9.

Domínguez Martínez, A., Cuetos Vega, F., 2018. The contrastive value of lexical stress in visual word recognition. Piscothema 30, 276–282.

Donselaar, W.v., Kostner, M., Cutler, A., 2005. Exploring the role of lexical stress in lexical recognition. The Quarterly journal of experimental psychology 58A, 251–273.

Fidelholtz, J., 1975. Word frequency and vowel reduction in English. Chicago Linguistics Society 11.

Friedrich, C.K., Kotz, S.A., Friederici, A.D., Alter, K., 2004. Pitch modulates lexical identification in spoken word recognition: ERP and behavioral evidence. Cognitive Brain Research 20, 300–308. doi:10.1162/089892904323057281.

Garcia, G.D., 2017. Weight gradience and stress in Portuguese. Phonology 34, 41–79.

Garcia, G.D., Guzzo, N.B., 2022. Lexical access in Portuguese stress. Journal of Portuguese Linguistics 21, 1–24. doi:10.16995/jpl.8188.

Goedemans, R., van der Hulst, H., 2009. StressTyp: A database for word accentual patterns in the world's languages, in: The Use of Databases in Cross-Linguistic Studies, pp. 235–282.

van Gompel, R.P.G., Hartsuiker, R.J., Kantola, L., 2026. The lexical boost in structural priming: The syntactic head matters but adjacency does not. Journal of Experimental Psychology: Learning, Memory, and Cognition Advance online publication. doi:10.1037/xlm0001599.

Gordon, M., 2002. A factorial typology of quantity-insensitive stress. Natural Language & Linguistic Theory 20, 491–552.

Guion, S.G., Clark, J.J., Harada, T., Wayland, R.P., 2003. Factors affecting stress placement for English nonwords include syllabic structure, lexical class, and stress patterns of phonologically similar words. Language and Speech 46, 403–427. doi:10.1177/00238309030460040301.

Gutiérrez-Palma, N., Palma-Reyes, A., 2007. On the use of lexical stress in reading Spanish. Reading and Writing 21, 645–660. doi:10.1007/s11145-007-9082-x.

Hartsuiker, R.J., Bernolet, S., Schoonbaert, S., Speybroeck, S., Vanderelst, D., 2008. Syntactic priming persists while the lexical boost decays: Evidence from written and spoken dialogue. Journal of Memory and Language 58, 214–238.

Hayes, B., 1983. A grid-based theory of English meter. Linguistic Inquiry 14.

Hayes, B., 1995. Metrical stress theory: principles and case studies. University of Chicago press.

Hilpert, M., 2008. The English comparative - language structure and language use. English Language and Linguistics 12, 395–417.

Huang, J., Liu, X., Lu, M., Sun, Y., Wang, S., Branigan, H.P., Pickering, M.J., 2023. The head constituent plays a key role in the lexical boost in syntactic priming. Journal of Memory and Language 131, 104416.

Jakobson, R., 1960. Linguistics and poetics, in: Sebeok, T.A. (Ed.), Style in language. MIT Press, Cambridge, MA, pp. 350–377.

Jusczyk, P.W., Cutler, A., Redanz, N.J., 1993. Infants' preference for the predominant stress patterns of English words. Child development 64, 675–687. doi:10.1111/j.1467-8624.1993.tb02935.x.

Kabak, B., Maniwa, K., Kazanina, N., 2010. Listeners use vowel harmony and word-final stress to spot nonsense words: A study of Turkish and French. Laboratory Phonology 1, 207–224.

Kantola, L., van Gompel, R.P.G., Wakeford, L.J., 2023. The head or the verb: Is the lexical boost restricted to the head verb? Journal of Memory and Language 129, 104388.

Ladd, D.R., 2008. Intonational phonology. 2nd ed., Cambridge University Press.

Lee, M.D., Wagenmakers, E.J., 2014. Bayesian cognitive modeling: A practical course. Cambridge University Press.

Levelt, W.J., 1993. Speaking: From intention to articulation. volume 1. MIT press.

Liberman, M., Prince, A., 1977. On stress and linguistic rhythm. Linguistic Inquiry 8, 249–336.

Martínez-Paricio, V., Kager, R., 2016. The binary-to-ternary rhythmic continuum in stress typology: layered feet and non-intervention constraints. Phonology 32. doi:10.1017/S0952675715000287.

McLean, J., Pickering, M.J., Branigan, H.P., 2004. Lexical repetition and syntactic priming in dialogue, in: Trueswell, J.C., Tanenhaus, M.K. (Eds.), Processing world-situated language: Bridging the language as product and language as action traditions. MIT Press, Cambridge, MA, pp. 193–208.

Moore-Cantwell, C., 2020. Weight and final vowels in the English stress system. Phonology 37, 657–695.

Moore-Cantwell, C., Bosch, D., Kahn, E.X., Kim, C., Shoemaker, G., 2024. Production priming of stress in nonwords. Laboratory Phonology 15, 1–31.

Morris, H., Keyser, S.J., 1971. Illustration and defense of a theory of the iambic pentameter. College English 33, 154–176.

van Oostendorp, M., 2015. Parameters in phonological analysis: stress, in: Fabregas, A., Putnam, M., Mateu, J. (Eds.), Contemporary linguistic parameters. Bloomsbury publishing. Contemporary studies in Linguistics, pp. 335–352.

Peperkamp, S., Vendelin, I., Dupoux, E., 2010. Perception of predictable stress: A cross-linguistic investigation. Journal of Phonetics 38, 422–430.

Pickering, M.J., Branigan, H.P., 1998. The representation of verbs: Evidence from syntactic priming in language production. Journal of Memory and Language 39, 633–651.

Pickering, M.J., Ferreira, V.S., 2008. Structural Priming: A Critical Review. Psychonomic Bulletin and Review 134, 427–459.

Pierrehumbert, J., 1980. The phonology and phonetics of English intonation. Ph.D. thesis. Massachusetts Institute of Technology.

Protopapas, A., Panagaki, E., Andrikopoulou, A., Gutiérrez Palma, N., Arvaniti, A., 2016. Priming stress patterns in word recognition. Journal of Experimental Psychology: Human Perception and Performance 42, 1739.

Reitter, D., Keller, F., Moore, J.D., 2011. A computational cognitive model of syntactic priming. Cognitive Science 35, 587–637.

Ryan, K., 2022. Degenerate feet in phrasal phonology: evidence from Latin and Ancient Greek. Phonology 39, 345–368.

Ryan, K.M., 2019. Prosodic weight: categories and continua. Number 3 in Oxford Studies in Phonology and Phonetics, Oxford University Press, Oxford, UK.

Scheepers, C., Raffray, C., Myachykov, A., 2017. The lexical boost effect is not diagnostic of lexically-specific syntactic representations. Journal of Memory and Language 92, 102–115. doi:10.1016/j.jml.2017.03.001.

Schild, U., Becker, A.B.C., Friedrich, C.K., 2014. Phoneme-free prosodic representations are involved in pre-lexical and lexical neurobiological mechanisms underlying spoken word processing. Brain and Language 136, 31–43.

Schiller, N.O., Fikkert, P., Levelt, C.C., 2004. Stress priming in picture naming: An SOA study. Brain and Language 90, 231–240.

Segaert, K., Wheeldon, L., Hagoort, P., 2016. Unifying structural priming effects on syntactic choices and timing of sentence generation. Journal of Memory and Language 91, 59–80.

Shaw, J., 2012. metrical rhythm in speech planning: priming or predictability, in: Proceedings of the 14th Australasian International Conference on Speech Science and Technology, Sydney, Australia. pp. 145–148.

Shaw, J.A., 2013. The phonetics of hyper-active feet: Effects of stress priming on speech planning and production. Laboratory Phonology 4, 159–190.

Shih, S.h., De Lacy, P., 2019. Evidence for sonority-driven stress. Catalan Journal of Linguistics 18, 9–40.

Shih, S.S., 2017. Phonological influences in syntactic choice, in: Gribanova, V., Shih, S.S. (Eds.), The morphosyntax-phonology connection: Locality and directionality at the interface. Oxford University Press, Oxford, pp. 223–252. doi:10.1093/acprof:oso/9780190210304.003.0009.

Snider, N., 2009. Similarity and structural priming, in: Taatgen, N.A., van Rijn, H. (Eds.), Proceedings of the 31st annual conference of the cognitive science society., Austin, Texas. pp. 816–822.

Soto-Faraco, S., Sebastián-Gallés, N., Cutler, A., 2001. Segmental and suprasegmental mismatch in lexical access. Journal of Memory and Language 45, 412–432. doi:10.1006/jmla.2000.2783.

Sulpizio, S., Arduino, L.S., Paizi, D., Burani, C., 2013. Stress assignment in reading Italian polysyllabic pseudowords. Journal of Experimental Psychology: Learning, Memory, and Cognition 39, 51–68.

Sulpizio, S., Vespignani, F., Job, R., 2016. On the time course of lexical stress priming in speech production: Behavioral and ERPs evidence from a free-stress language. Brain Research doi:http://dx.doi.org/10.1016/j.brainres.2016.07.018.

Tabossi, P., Tagliapietra, L., 2005. Lexical stres effects in Italian spoken word recognition, in: Proceedings of the Annual Meeting of the Cognitive Science Society, Italy, 27, pp. 2140–2144.

core team, R., 2025. R: A language and environment for statistical computing. URL: https://www.R-project.org/.

Thiessen, E.D., Saffran, J.R., 2003. When cues collide: use of stress and statistical cues to word boundaries by 7- to 9-month-old infants. Developmental psychology 39, 706–716.

Tooley, K.M., 2023. Structural priming during comprehension: A pattern from many pieces. Psychonomic Bulletin and Review 30, 882–896.

Zehr, J., Schwarz, F., 2018. PennController for Internet Based Experiments (IBEX). doi:10.17605/OSF.IO/MD832.